

Script-Controlled ACL – Restrict Record Access Based on Field Value

5. Project Development Phase

The development phase implements the planned record-level security in ServiceNow. The main development component is the creation of the **u_institution_details** table, the required user roles, and script-controlled Access Control Lists for Read, Create, Write, and Delete operations.

The Institution Details table is created under System Definition → Tables with the required fields including Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, and Description. The Branch field is configured with the choices **ECE, EEE, and CSE**. Test records are then created with different branch values to verify the access restrictions.

The Read ACL is developed for the **u_institution_details** table with the operation set to Read and Advanced enabled. The **bb1** role is added to the Requires role section, and the data condition is configured as **Branch is EEE**. The script allows administrators full access, allows users with the required role, and denies access to users who do not satisfy the configured conditions.

Separate ACLs are then developed for the remaining operations. The Create ACL requires **bb2**, the Write ACL requires **bb3**, and the Delete ACL requires **bb4**. These ACLs are configured for the same table and saved after the required role settings are applied.

After the ACLs are configured, the implementation is tested using impersonation. The EEE user is used to verify access to permitted EEE records and the availability of the required operations. Users without the required roles are tested to confirm that access is denied, while the administrator account is tested to confirm full access to all Institution Details records.